

Bharat Yalavarthi

636-891-5679 | bharatcy2@gmail.com | [linkedin.com/in/bharat-yalavarthi](https://www.linkedin.com/in/bharat-yalavarthi) | byalavar.github.io

RESEARCH INTEREST

Interpreting deep learning models, and using interpretability techniques for alignment, robustness and diagnosing model failure modes.

EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

Ph.D. in Computer Science, GPA: 3.8/4.0, UB Presidential Fellowship

Jan 2025 - May 2027 (expected)

Advisors: Dr. Nalini Ratha, Dr. Venu Govindaraju

M.S. in Computer Science, GPA: 3.9/4.0

Aug 2022 - May 2024

SELECTED PUBLICATIONS

Full list on [Google Scholar](#)

- **Label-Free Mitigation of Spurious Correlations in VLMs Using Sparse Autoencoders** ICLR 2026
B. C. Yalavarthi, N. Ratha, V. Govindaraju
- **Aligning Characteristic Descriptors with Images for Human-Expert-like Explainability** NeurIPS 2024 IAI Workshop
B. C. Yalavarthi, N. Ratha
- **Enhancing Privacy in Face Analytics Using Fully Homomorphic Encryption** IEEE FG 2024
B. C. Yalavarthi, A. R. Kaushik, A. Ross, V. Boddeti, N. Ratha

RESEARCH EXPERIENCE

University at Buffalo

Jan 2025 - Present

PhD Researcher

Buffalo, NY

- **(ICLR 2026)** Developed DIAL, a label-free Sparse Autoencoder (SAE) method that removes spurious correlations from VLM representations. Improved worst-group accuracy by up to 37 points across 5 standard benchmarks without labels, extra data, or retraining [\[Paper\]](#) [\[Code\]](#).
- Applied entropy regularization to Concept Bottleneck Models (CBMs), cutting information leakage by 40% (under review).
- Probed concept formation across Gemma Scope SAE scales: wide SAEs tend to form compositional feature families (not atomic features), while narrow SAEs tend to collapse frequent concepts into coarse superpositions.

University at Buffalo (SUNY Research Foundation)

Aug 2024 - Jan 2025

Researcher

Buffalo, NY

- **Benchmark Creation:** Designed a Gestalt-grounded benchmark (1,200+ stimuli, 4 perceptual principles) revealing systematic visual reasoning failures in GPT and Claude models that standard VQA benchmarks miss.
- Stress-tested SOTA MLLMs on OCR through targeted perturbations, isolating regimes where vision encoders break down.

University at Buffalo (SUNY Research Foundation)

Jan 2023 - May 2024

Research Assistant

Buffalo, NY

- **Explainability:** Used Mistral-7B and CLIP to generate human-expert-like justifications for face recognition and medical predictions, improving explanation faithfulness through characteristic descriptors (NeurIPS 2024 IAI Workshop).
- **Privacy & Secure ML:** Exposed attribute leakage in face-template protection schemes and built an FHE-based mitigation that cut leakage by 35% over prior methods with no accuracy loss (IEEE FG 2024, 2025).

INDUSTRY EXPERIENCE

Harman International (Samsung)

Oct 2020 - Jul 2022

Software Engineer, ADAS & Navigation

Bangalore, India

- Delivered 20+ ADAS and route guidance features for VW Trucks and Coaches, with a modular redesign that reduced software maintenance costs by 25%.
- Led the product line expansion of ADAS modules and integrated an onboard traffic sign recognition module, cutting incorrect sign events by 90%.

SKILLS & ACADEMIC SERVICE

- **Skills:** Mechanistic interpretability (SAEs, activation patching, causal tracing, probing, transcoders), evaluation and benchmark design, red-teaming, PyTorch, Hugging Face, SAELens, Gemma Scope, NumPy, DDP, Docker, large language models (LLMs), vision-language models (VLMs), Python, C++
- **Reviewer:** NeurIPS, CVPR, WACV. Presented at NSF CITeR Program Review 2024.